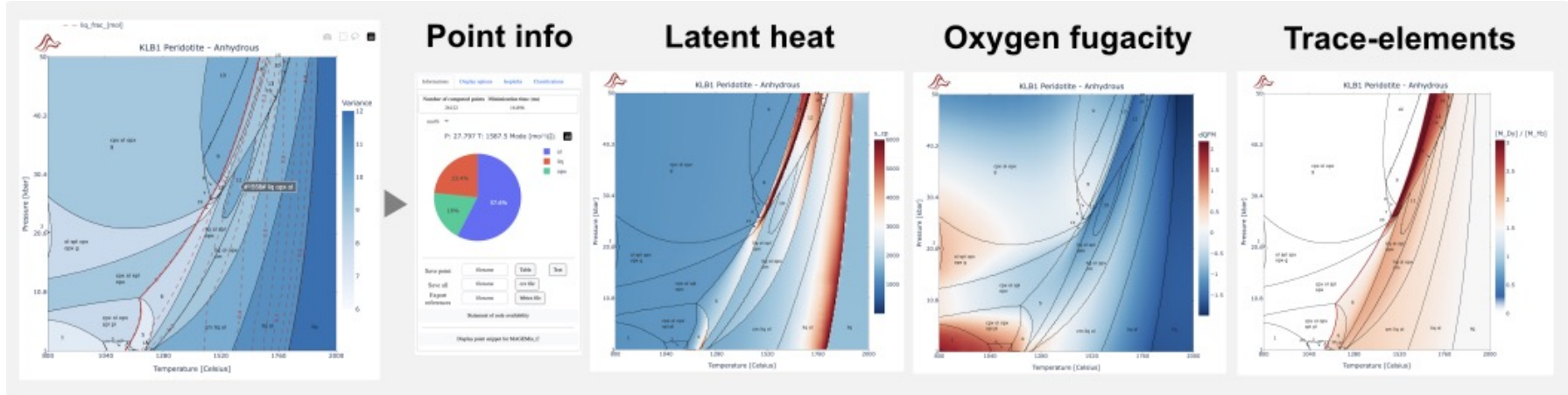
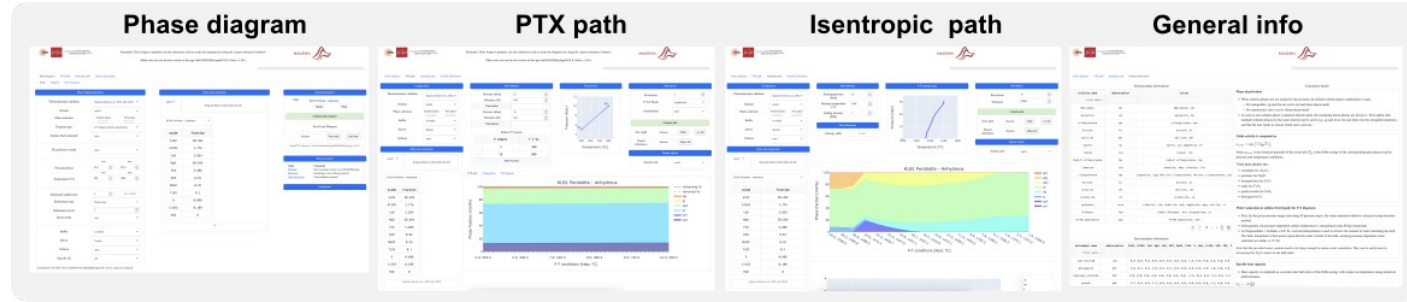


Phase Equilibrium Modelling with MAGEMin

Nicolas Riel - Johannes-Gutenberg University Mainz, Germany nriel@uni-mainz.de

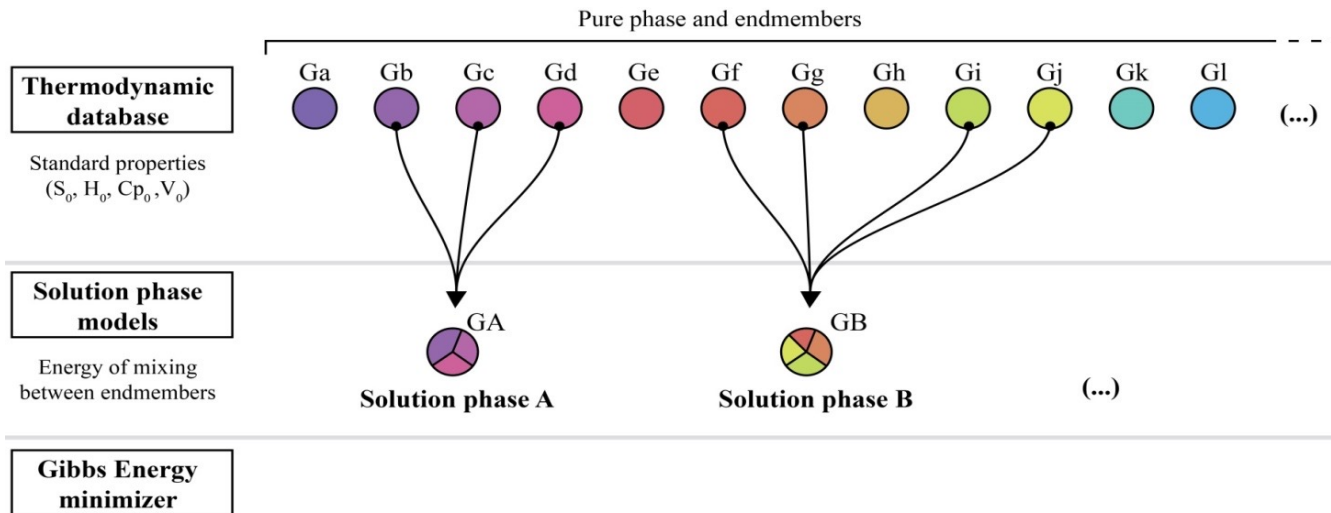


Overview



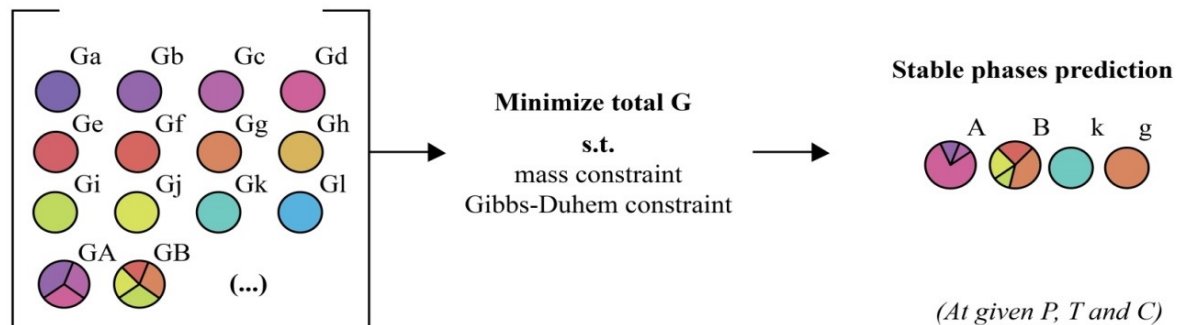
- General introduction and rationales
- Current/ongoing state of development
- Useful links (*repositories, documentation and discord*)

Phase equilibrium modeling: components

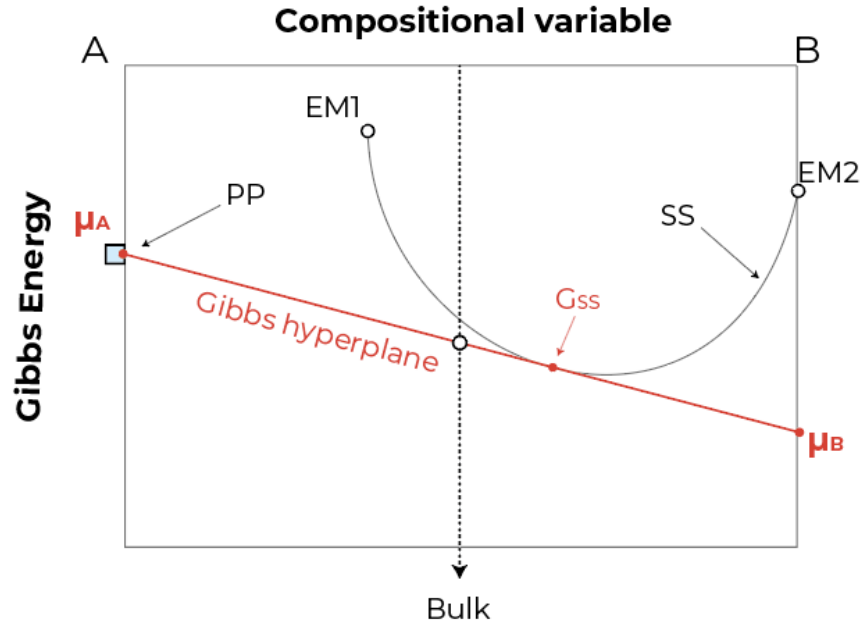


e.g.,
tc-ds6xx.txt for TC
hp02ver.dat for Perple_X

e.g.,
a-x models for TC



Phase equilibrium modeling: anatomy of an equilibrium



At fixed P-T conditions

$$\text{Bulk} = \alpha C_{PP} + (1 - \alpha) C_{SS} \quad \left| \quad \text{Mass constraint} \right.$$

$$G_{SS} = C_{A_{SS}} \mu_A + C_{B_{SS}} \mu_B$$

$$G_{PP} = C_{A_{PP}} \mu_A + C_{B_{PP}} \mu_B$$

Gibbs-Duhem constraint

Why a new Gibbs free energy minimizer?

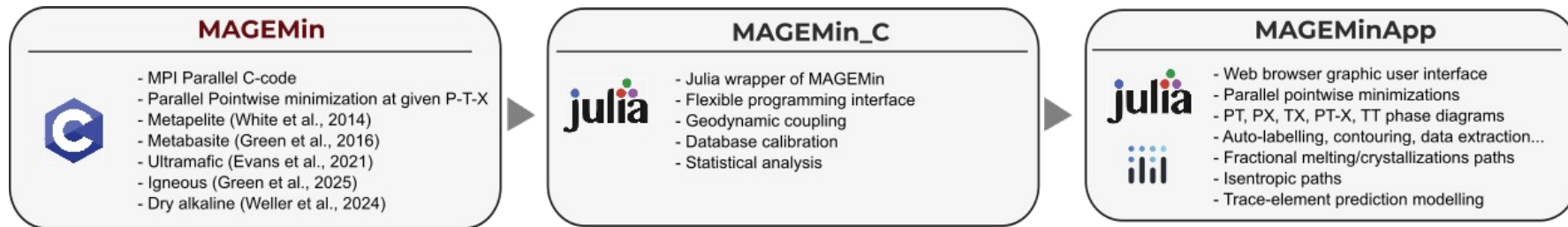
Perple_**x**



MELTS

- Developed (in the 90's) to produce phase diagrams
- Natively not parallel
- Minimizations can often fail
- Not optimized for solution update
- Difficult to couple to geodynamic codes (written in Fortran)

MAGEMin: current stage and ongoing developments



Current stage

- MPI parallel C-library with several built-in database
 - Holland et al., 2013; White et al., 2014; Green et al., 2016; Evans & Forst, 2021; Weller et al., 2024; Green et al., 2025; Stixrude & Lithgow-Bertelloni 2011, 2021;
- Benchmarked 1 to 1 against THERMOCALC (for TC database)
- Julia wrapper for petrological/geodynamic coupling (MAGEMin_C.jl)
- Parallel web-browser Julia app (MAGEMinApp.jl)
- Single core performances
 - ~10 to 150 ms without initial guess (scales with #oxides and #solutions phases)
 - ~1 to 20 ms with initial guess



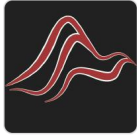
Ongoing development

- Algorithm upgrade to improve performances and stability (Riel et al., 2025, accepted)
- Inversion framework → AutomaticDifferentiation → rewriting MAGEMin in pure Julia

MAGEMin repository



<https://github.com/ComputationalThermodynamics>

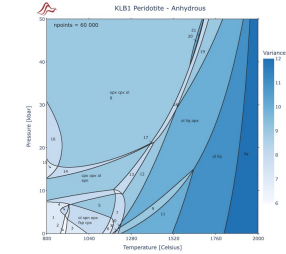


ComputationalThermodynamics

Tools for thermodynamic computing

👤 28 followers

📍 Germany



nriel@uni-mainz.de

Popular repositories

MAGEMin

Public

The parallel Mineral Assemblage Gibbs Energy Minimization package

● C ☆ 86 🍴 20

MAGEMinApp.jl

Public

Graphical User Interface for MAGEMin, which runs in your web-browser.

● Julia ☆ 16 🍴 3

MAGEMin_C.jl

Public

Julia interface to the MAGEMin C package

● C ☆ 14 🍴 5

Resources

Public

● Julia ☆ 2



MAGEMin documentation



https://computationalthermodynamics.github.io/MAGEMin_C.jl/dev/

MAGEMin Toolset

Phase equilibrium modelling

A set of tools to compute phase equilibria for Earth-like compositions



Getting Started

MAGEMin_C.jl API

View on GitHub



MAGEMinApp

Effortlessly compute phase diagrams and Pressure-Temperature paths



MAGEMin_C

Set of function to freely compute phase equilibrium in Julia



MAGEMin

C program backend of MAGEMin

MAGEMinApp.jl

Introduction

Installation

Tutorials - Bulk input file

Tutorials - Phase diagrams

Tutorials - P-T-X paths

Tutorials - Isentropic paths

Citation

Interface

MAGEMin_C.jl

Introduction

Installation

Tutorials - Output

Tutorials - Step by step

Tutorials - Quick examples

API



MAGEMin discord channel



<https://discord.gg/fjmVZyej9F>



- **Report bugs!**
- **If functions/options are missing don't hesitate to ask for addition!**
- **Everything is open source: you can contribute by creating pull-requests on GitHub!**

Workshop program

- **Installation**
- **Introduction of MAGEMinApp interface**
 - Functionality
 - Post-processing
 - Data export
- **Exercises (set of MAGEMinApp online tutorials)**
- **Introduction of MAGEMin_C programming interface**
 - Functionality
 - Output structure
- **Exercises (set of MAGEMin_C online tutorials)**

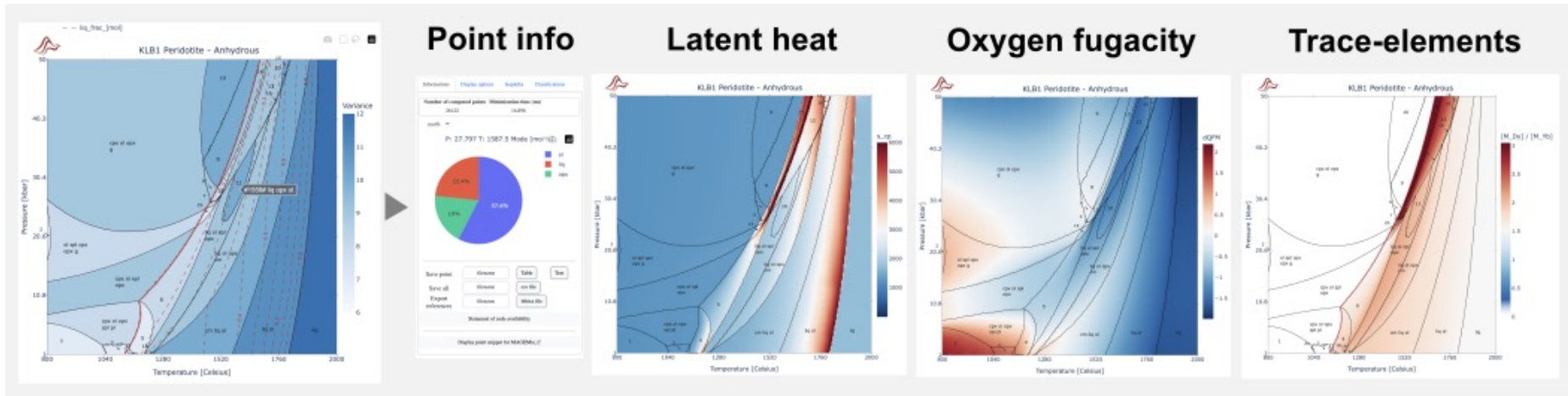


MAGEMinApp



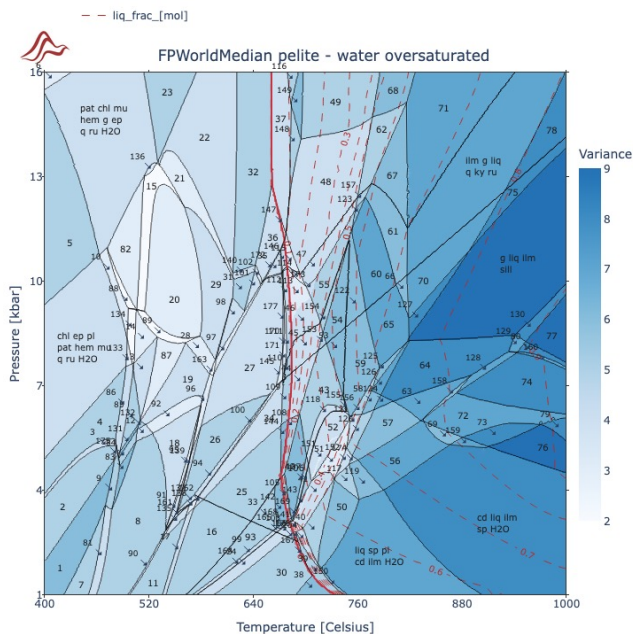
MAGEMin_C

MAGEMinApp.jl

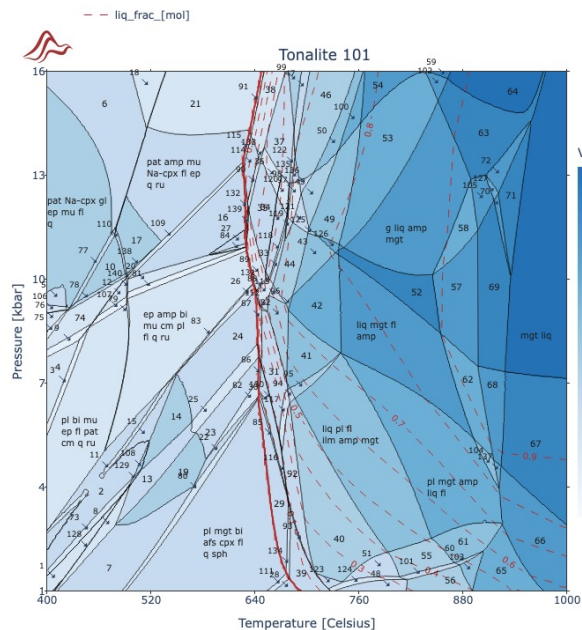


- P-T, T-X, P-X, PT-X, T-T phase diagrams
- Trace-element partitioning
- PTX paths (fractional melting/crystallization)
- Automatic labelling, isocontours, vector format figures, export data as `.csv` etc.

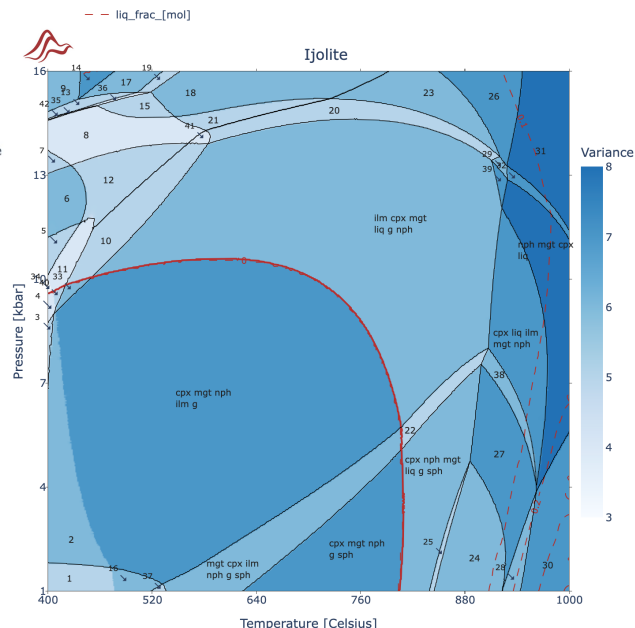
MAGEMinApp: examples of phase diagrams



- World Median metapelite
- Metapelite – White et al., 2014



- Tonalite 101
- Igneous – Green et al., 2025



- Ijolite
- Alkaline dry – Weller et al., 2024

Workshop program

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MAGEMinApp

MAGEMin_C

MAGEMin_C.jl

- Programmable Julia interface to MAGEMin
- Parallel calculations (multi-threaded)
- Easy to use

```
using MAGEMin_C
data = Initialize_MAGEMin("ig", verbose=false);
P,T = 10.0, 1100.0
Xoxides = ["SiO2"; "Al2O3"; "CaO"; "MgO"; "FeO"; "Fe2O3"; "K2O"; "Na2O"; "TiO2"; "Cr2O3"; "H2O"];
X = [48.43; 15.19; 11.57; 10.13; 6.65; 1.64; 0.59; 1.87; 0.68; 0.0; 3.0];
sys_in = "wt"
out = single_point_minimization(P, T, data, X=X, Xoxides=Xoxides, sys_in=sys_in)
```

- Stores all output in a structure

```
out.
G_system Gamma MAGEMin_ver M_sys PP_vec P_kbar SS_vec T_C V Vp Vp_S Vs Vs_S X
aAl2O3 aFeO aH2O aMgO aSiO2 aTiO2 alpha bulk bulkMod bulkModulus_M bulkModulus_S bulk_F bulk_F_wt bulk_M
bulk_M_wt bulk_S bulk_S_wt bulk_res_norm bulk_wt cp dQFM dataset enthalpy entropy fO2 frac_F frac_F_wt frac_M
frac_M_wt frac_S frac_S_wt iter mSS_vec n_PP n_SS n_mSS oxides ph ph_frac ph_frac_vol ph_frac_wt ph_id
ph_type rho rho_F rho_M rho_S s_cp shearMod shearModulus_S status time_ms
```